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Project #17: Addressing Safety Issues with Direct Oral Anticoagulants

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AIM

Problem: Over the past decade, direct oral anticoagulants (DOACs) have become the medication of choice for treatment of atrial fibrillation, thrombosis, and some coronary artery disease states. Prescribing these medications is often associated with inaccurate dosing leading to increased risk of bleeding and clotting events.

Events related to DOAC medications have led the Joint Commission to create a National Patient Safety Goal as of July 2019, aiming to reduce the likelihood of patient harm associated with the use of anticoagulant therapy.¹

A meta-analysis of 18 DOAC studies found that on average, prescribing errors or inappropriate dosing occurred in 20% of patients.² These errors can lead to bleeding, clotting and overall increase in mortality.³ Currently monitoring for dosing errors is limited at Henry Ford Health.



Quality Improvement

Plan: To provide improved quality of oversight of direct oral anticoagulants.

Henry Ford Health is part of the Michigan Anticoagulation Quality Improvement Initiative (MAQI²) funded by Blue Cross Blue Shield of Michigan. This initiative helped to develop the DOAC Dashboard. This is a tool modeled after the Veterans Health Administration to assist with monitoring potential safety issues surrounding DOAC Medications⁴

Henry Ford Medical Group Ambulatory Anticoagulation Services (ACS) monitors warfarin and DOAC anticoagulants. The DOAC Dashboard seems to be a perfect fit for the efforts of the ACS team. This Population Management Tool will aid in improved provider prescribing and elevated patient safety by ensuring patients receive the correct doses of these high alert medications. The DOAC Dashboard tool is a system set up in EPIC and can identify safety alerts surrounding dosing and other potential alerts with these medications. The dashboard breaks them down into categories of known critical alert, possible critical alerts and FYI alerts.



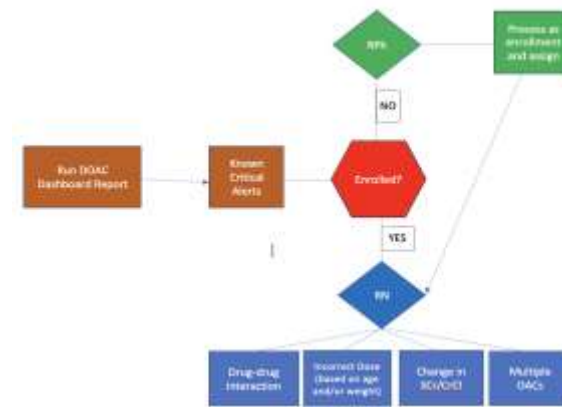
Evaluation

After reviewing the type of alerts the system provides, we decided to start with “known critical alerts.” These alerts pose the greatest safety issues. In addition, we further reduced the number of patients by just choosing patients established in the Henry Ford Medical Group. This was chiefly because the ACS already manages anticoagulation therapies for this group of physicians and patients, and the amount of staffing available. A pilot for this program was done by two nurses with anticoagulation experience who tested the program for 6 months. Based on time and staffing this was approximately 8 hours a week. During this time, they were able to review, manage and correct 358 critical alerts that patients had with inaccurate DOAC dosing.

Action/Intervention

The HFMC Ambulatory Anticoagulation Service (ACS) is comprised of pharmacists and nurses who have been specially trained to be experts in anticoagulation therapy management. This HFMC ACS team had been formally practicing patient DOAC therapy management for over a decade by educating patients newly started on a DOAC and following up with their care on a scheduled basis.

Before DOAC Dashboard use was implemented into the HFMG ACS workflow, a small HFMG ACS pilot group was trained to use the tool so it could be tested before it was rolled out to the rest of the team. During an educational teleconference, the functionality and process was presented to the rest of the HFMG ACS team prior to officially integrating the DOAC Dashboard into daily patient care operations.



Photos

Example of alert in EPIC

Known critical alert - Incorrect dose - Age: 80; weight: 114.76 kg; Creatinine: 3.10 mg/dL

DOAC: apixaban	Current dose: 5mg BID	Current medication directions: Take 1 tablet (5 mg total) by mouth (two) times daily.
Indication: Non-valvular atrial fibrillation/flutter -- from Problem List/History	Expected dose: 2.5mg BID	Drug-drug interactions: N/A

	Latest Ref	5/15/2023	5/14/2023	5/13/2023	5/12/2023	5/12/2023	5/11/2023	5/11/2023
	Rng & Units	8:25 AM	7:35 AM	6:11 AM	5:47 PM	3:58 PM	10:25 AM	10:11 AM
Creatinine and Weight								
Creatinine	<1.13 mg/dL	3.10	1.20	1.01	1.33		1.53	
Weight (kg)						114.76		113.399

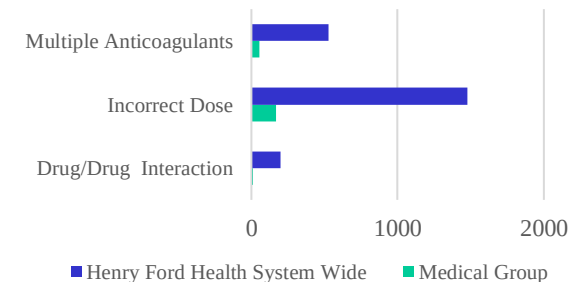
Measures

The DOAC dashboard transition to the ACS has been a steady but slow process. On average it takes a pharmacist or nurse 20 minutes to complete an alert on the Dashboard, which is in line with the time required to manage other anticoagulation patient care. We are working with MAQI² to help develop measurement tools for the system to keep track of time spent addressing and completing alerts. This will help to project staff timing and funding to ensure the dashboard continues.

The DOAC Dashboard is an aid to ensure patients are dosed appropriately per manufacturer recommendations and evidence-based guidance, thus reducing patient risk for bleeding, thrombosis and over all-mortality. In 2024 we will hopefully be able to measure the impact the Anticoagulation Service impact has been on the reduction of DOAC medication errors and inappropriate dosing.

Spread

The ACS continues to work on known critical alerts on the DOAC dashboard with plans to slowly move to other alerts. These alerts currently are only patients in the Henry Ford Medical Group about 6,000 patients. This is only a fraction of the patients in Henry Ford Health. The DOAC Dashboard system has the ability to reach the entire Henry Ford Health network which includes over 29,000 patients who have been prescribed a DOAC medication.



Keys to Success

- Dedication of the Henry Ford Medical Group Ambulatory Anticoagulation Services (ACS)
- Being part of the Michigan Anticoagulant Quality Improvement Initiative (MAQI²), which is comprised of administration, faculty and staff dedicated to improve safety with DOAC medication
- Funding from Blue Cross Blue Shield of Michigan.

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